Graph Coloring

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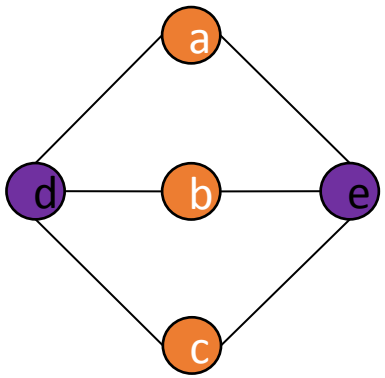
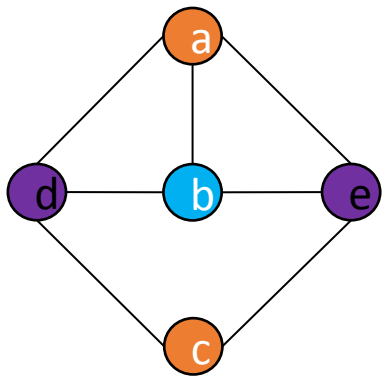
Graph Coloring

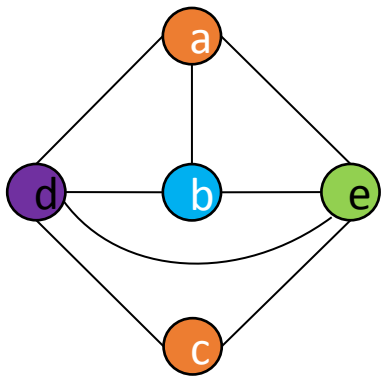
- **What** is the **minimum number of colors required** to **paint all** the **vertices** of **graph G** with **n** vertices such **that no two adjacent vertices** have the **same color**? Here **graph G** is **simple** and **connected**.
- This **coloring allows grouping** of **vertices** into **multiple partitions**(**sets**). **Note**: **spanning tree** is about **partitioning** the **edges** of a given graph.



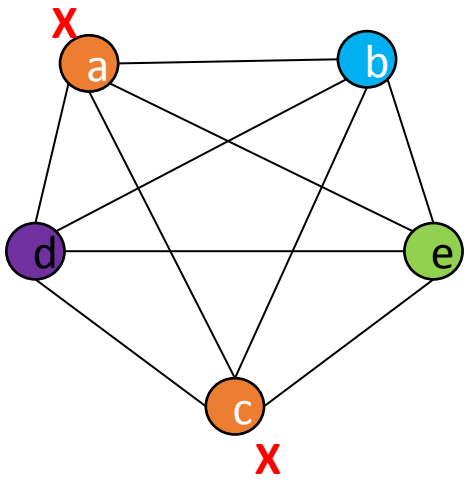
minimum number of colors required =2



minimum number of colors required =3



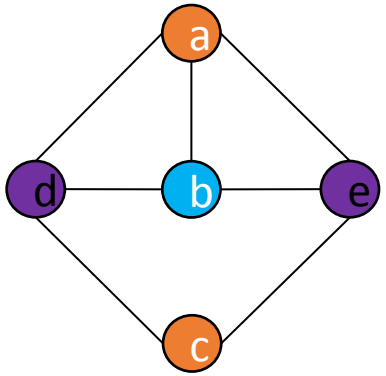
minimum number of colors required =4



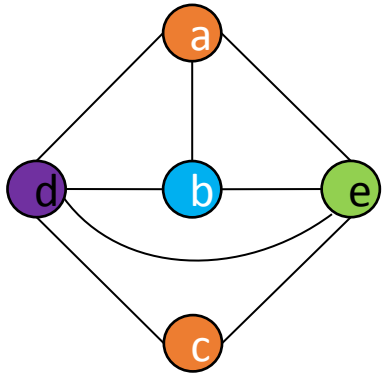
4 colors not sufficient for this graph, need minimum 5.

Graph Coloring

- **Chromatic Number(k)** is the **minimum number of colors required** to **paint all** the **vertices** of **graph G** with **n** vertices such **that no two adjacent vertices** have the **same color**. The graph with **chromatic Number k** is called a **k-chromatic** graph.
- A **graph** consisting of **only isolated vertices** is **1-chromatic**.
- A **complete graph of n vertices** is **n-chromatic** (as all its vertices are adjacent).
- A graph **containing** a **complete graph** of **r vertices** is **at least r-chromatic**. For e.g. **graph G** having **triangles** is **3-chromatic**.



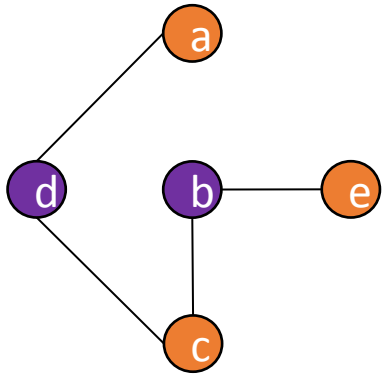
Graph G
k=3



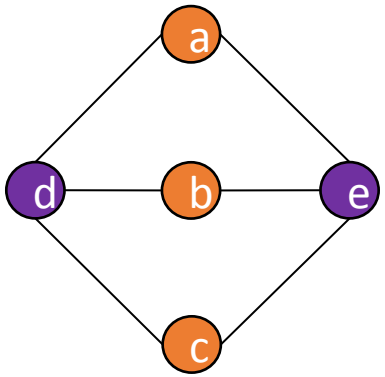
Graph H
k=4

Graph Coloring

- Every **tree** with **two** or **more vertices** is **2-chromatic**.
  - Starting from the root, we can color nodes in successive levels using alternate colors. Therefore **2 colors** are **sufficient** to **color** a **tree**.
- But **every 2-chromatic graph** is **not a tree**.



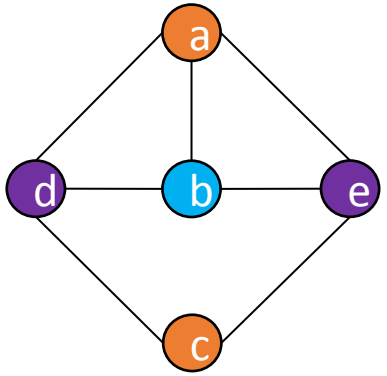
Tree T, k=2



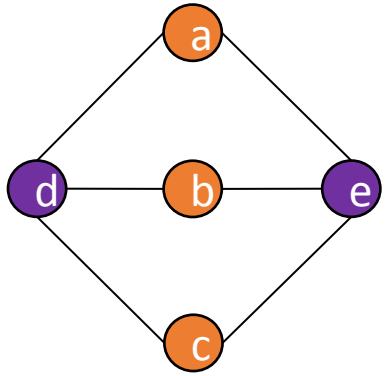
Not a tree but k =2

Graph Coloring

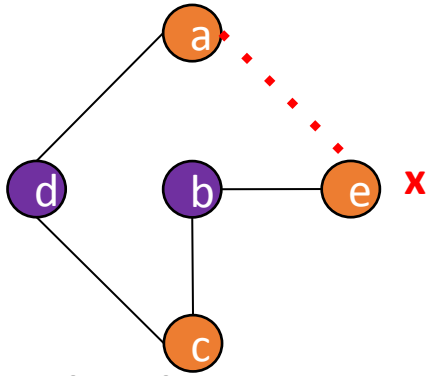
- A **graph** with **at least one edge** is **2-chromatic if and only if** it has **no circuits of odd length**.



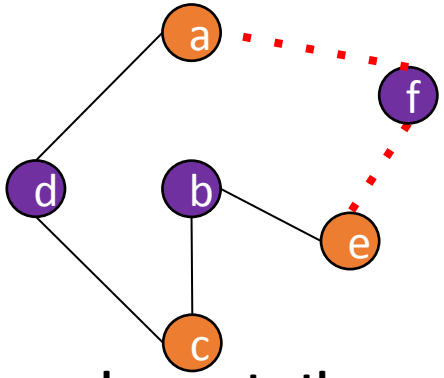
Graph contains circuits of length 3
minimum number of colors required =3



Graph don't have circuits of odd length
2-chromatic graph (minimum number of colors required =2)



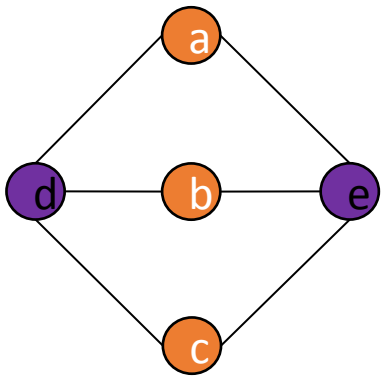
Addition of a cord to the spanning tree create a circuits of odd length; graph is not 2-chromatic.



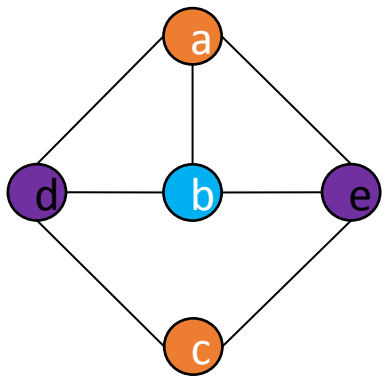
Addition of cords one by one to the spanning tree create a circuits of even length; graph is 2-chromatic.

Graph Coloring

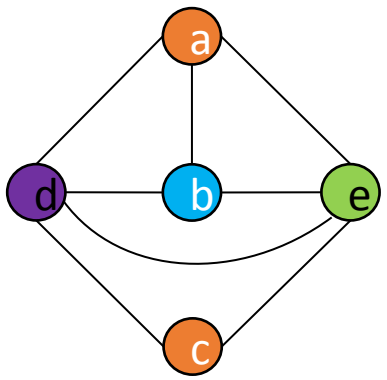
- If d_{max} is the **maximum degree** of the vertices in a **graph G**, **chromatic number k** of **G** $\leq 1 + d_{max}$.



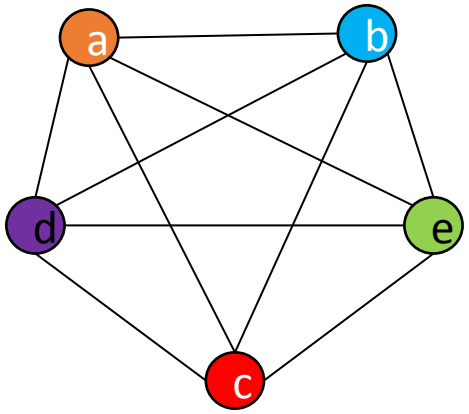
d_{max} = **3** chromatic number = **2**



d_{max} = **3** chromatic number =**3**



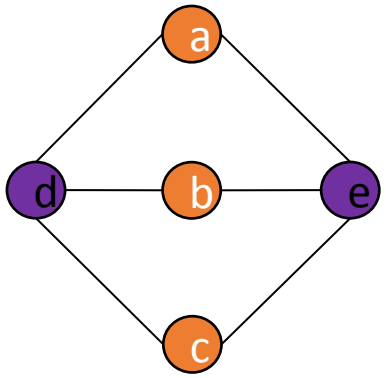
d_{max} = **4** chromatic number = **4**



d_{max} = **4** chromatic number = **5**

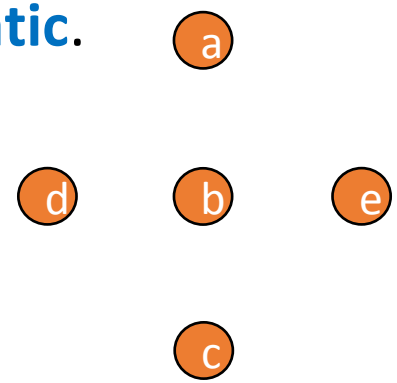
Graph Coloring

- Every **2-chromatic graph** is **bipartite** because the coloring partitions the vertex set into two subsets V1 and V2 such that no two vertices in V1 (or V2) are adjacent.



bipartite graph, chromatic number = **2**

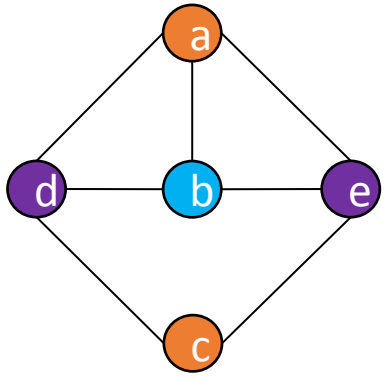
- Every **bipartite** graph is **2-chromatic**, with **one trivial exception**
  - a **graph** of **two or more isolated vertices** (and with no edges) is **bipartite** but is **1-chromatic**.



bipartite graph, **Set 1 contains all the isolated nodes**, and **set 2** is **empty**.
v1={a, b, c, d, e}, **v2**= { }, chromatic number = **1**

Independent Set

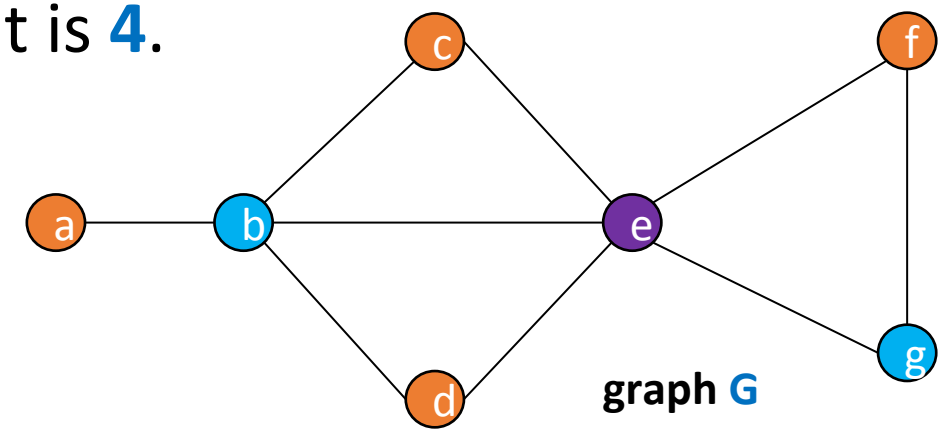
- **Coloring** of a **graph** induces a **partitioning of the vertices** into different **subsets**. For example, the coloring in graph **G** produces the partitions
  - **{a, c}**, **{b}**, and **{d, e}**.
- **No two vertices in** any of these three **subsets** are **adjacent**. Such a **subset of vertices** is called an **independent set**.
- A set of vertices in a graph is said to be an independent set (or an internally stable set) if no two vertices in the set are adjacent.
- A **single vertex** in any graph **constitutes** an **independent set**. Because, that single vertex makes a set and there is no scope of adjacent vertex in that set.



graph **G**

Maximal Independent Set

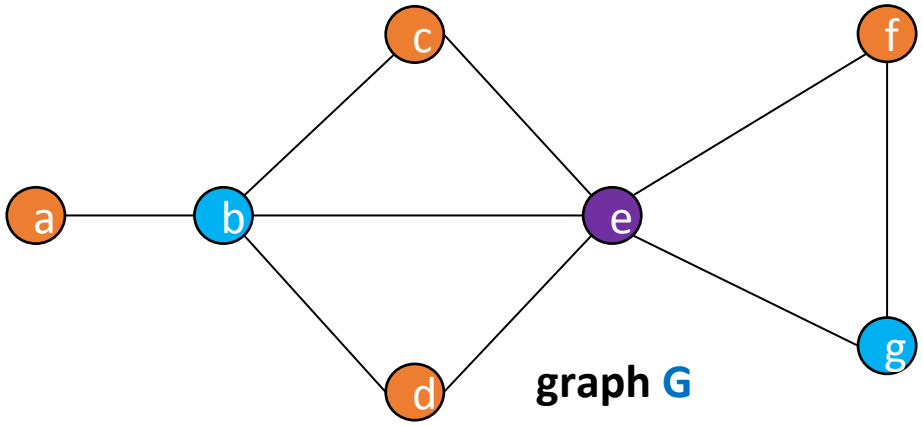
- A **maximal independent set** (or maximal internally stable set) is **an independent set** to which **no other vertex can be added without destroying** its **independence property**.
- The sets **{a, c, d, f}**, **{b, f}**, **{b, g}** in graph **G** are **maximal independent sets**. **A graph**, in general, **can have many maximal independent sets**; and they may be **of different sizes**. **Vertices** of a **set need not be** of **same color**.
- The **number of vertices in the largest (maximal) independent set** of a graph *G* is called the **independence number** (or **coefficient of internal stability**), **β (G)**. For **graph G** it is **4**.



graph G

Maximal Independent Set

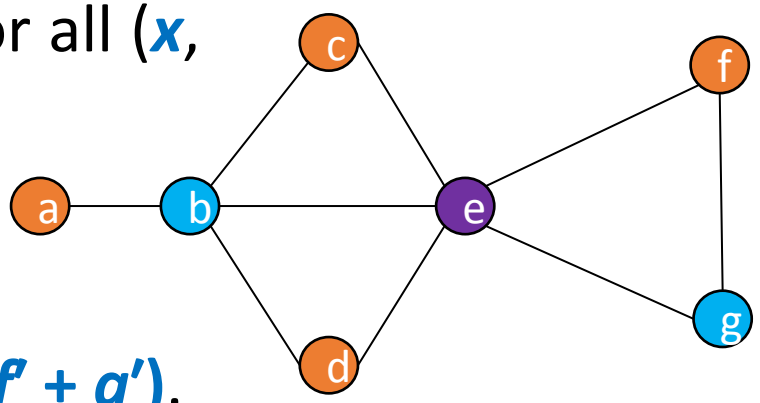
- The **independence number β(G)** for **graph G** it is **4**. The chromatic number **k** for **G** is **3**.
- **largest number** of vertices with the **same color** in **G cannot exceed** the **independence number β(G)**. Therefore for a **graph with n vertices**, **β(G) >= n/k**. For graph **G**, **β(G) = 4(>=7/3)**
- **Finding a Maximal Independent Set**:
  - **start** with **any vertex *v*** of ***G*** in the set.
  - **add more vertices** to the set, **selecting** at each stage a **vertex** that is **not adjacent** to any of those **already selected**.
- This procedure will ultimately produce a maximal independent set. This set, however, is not necessarily a maximal independent set with a largest number of vertices.



graph G

Finding all Maximal Independent Set

- Let an **edge** (x, y) is **expressed** as a **Boolean product**, xy and **sum** of **all such products** in G as a Boolean expression $\varphi = \Sigma xy$ for all (x, y) in G.
  - For example, in **G**, $\varphi = ab + bc + bd + be + ce + de + ef + eg + fg$
- Let the **Boolean complement** φ' of this expression φ be
- $\varphi' = (a'+b')(b' + c')(b' + d')(b'+e')(c'+e')(d'+e')(e'+f')(e' + g')(f' + g')$.
- Expressed φ' as a **sum of Boolean products** $\varphi' = f_1 + f_2 + \ldots + f_k$ by simplifying as $\varphi' = b'e'f' + b'e'g' + a'c'd'e'f' + a'c'd'e'g' + b'c'd'f'g'$

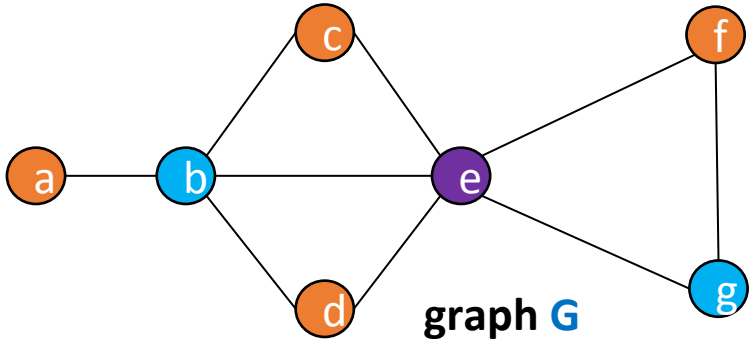


graph G

- The set of vertices, whose complements are not present in a term in φ' is a **Maximal Independent Set**. **For example**, vertices **a**, **c**, **d**, and **g** are **not present** in the term $b'e'f'$ and therefore **{a, c, d, g}** is a **Maximal Independent Set**.
- Considering the **remaining terms**, the other **Maximal Independent Sets** can be found as **{a, c, d, f}**, **{b, g}**, **{b, f}**, and **{a, e}**

Finding Independence & Chromatic Numbers

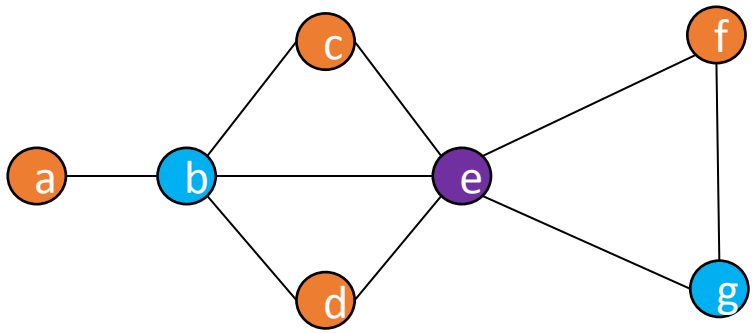
- The **size** of the **maximal independent set** with the **largest number of vertices** is the **independence number $\beta(G)$**.
- The maximum size among the set of {**a, c, d, g**}, {**a, c, d, f**}, {**b, g**}, {**b, f**}, and {**a, e**} is **4**, therefore, independence number of the **graph G** is **four**.
- To find **k**, the **chromatic number** of a graph G, **is equal to** the **minimum number of maximal independent sets**, which **collectively include all the vertices** of **G**.
- For the **graph G**, sets {**a**, **c**, **d**, **f**}, {**b**, **g**}, and {**a**, **e**}, for example, satisfy this condition. Thus the graph is **3-chromatic**.



graph G

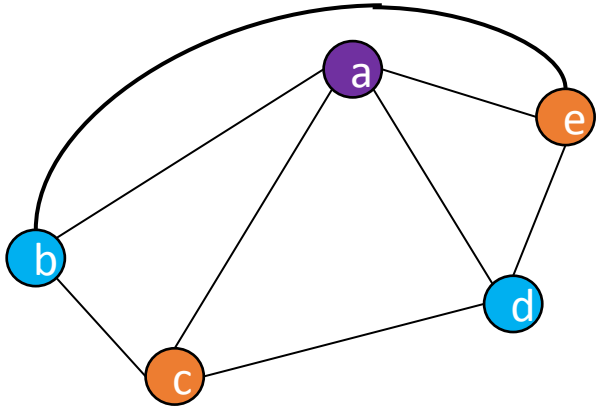
Chromatic Partitioning

- **Chromatic Partitioning**: Given a simple, connected graph G, **partition all vertices** of **G into** the **smallest possible number of disjoint**, **independent sets**.
- **Procedure**: enumerating **all maximal independent sets** and then **selecting** the **smallest number of sets that include all vertices** of the graph. **For example**, some of the **chromatic partitions** of the **graph G** are **{**(*a*, *c*, *d*, *f*), (*b*, *g*), (*e*)**}**, **{**(*a*, *c*, *d*, *g*), (*b*, *f*), (*e*)**}**, **{**(*c*, *d*, *f*), (*b*, *g*), (*a*, *e*)**}**, **{**(*c*, *d*, *g*), (*b*, *f*), (*a*, *e*)**}**.
- **Uniquely Colorable Graphs**: A graph having **only one chromatic partition** is called a uniquely colorable graph. **Graph G** is **not Uniquely Colorable** whereas **Graph H** is **uniquely colorable**, as the only **chromatic partition is {**(*a*), (*b*, *d*), (*c*, *e*)**}**.



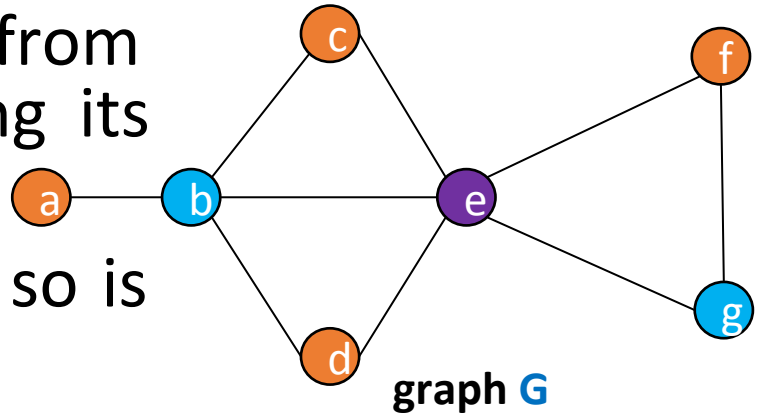
graph G

graph H



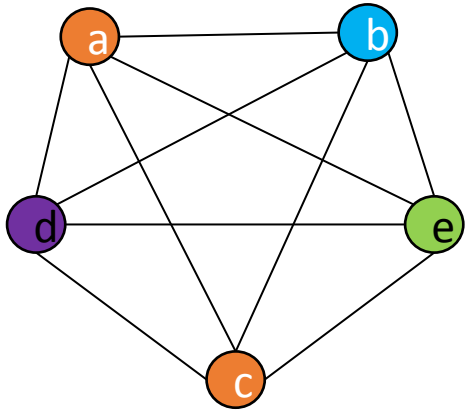
Dominating Sets

- **Dominating Vertex**: A **vertex v** in a **Graph G** is said to **dominate itself** and also its **neighbors**.
- **Dominating Set**: A **set S** of **vertices** of a **Graph G** is said to be **dominating set if every vertex** of **G** is **dominated by some vertex in S**. For example, **{***b, g***}** as these two vertices are sufficient to dominate all the vertices. The **other examples** are sets **{***b, e***}**, **{***b, f***}**, **{***a,* c, d, f**}** etc.
- The set of **all vertices {***a, b,* c, d, e, f, *g***}** is **trivially** a **dominating set** in the graph G.
- **Minimal Dominating Vertex**: is a dominating set from which no vertex can be removed without destroying its dominance property.
- For example, **{**b, e**}** is a minimal dominating set, and so is **{**a, c, d, f**}**.



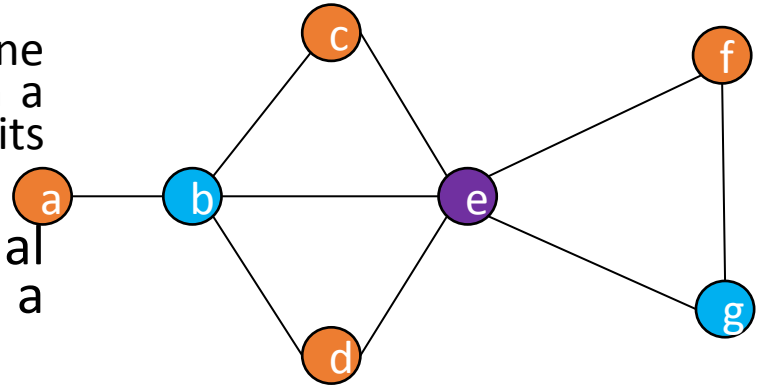
Minimal Dominating Set

- **Any one vertex** in a **complete graph** constitutes a **minimal dominating set**(e.g. **Graph G**).
- Every **dominating set contains at least one minimal dominating set**.
- A **graph** may have **many minimal dominating sets**, and of **different sizes**. The **examples** in **graph H** are sets **{***b, g***}**, **{***b, e***}**, **{***b, f***}**, **{***a,* c, d, f**}**
- The **number of vertices in the smallest minimal dominating set** of a graph *G* is called the ***domination number*, *α*(G)**.
- **Minimal dominating set may or may not be independent**. **For example**, **{***b, g***}** is a **independent set**, but **{***b, e***}** is **not**.
- Every **maximal independent set is a dominating set**.
  - For if an independent set does not dominate the graph, there is at least one vertex that is neither in the set nor adjacent to any vertex in the set. Such a vertex can be added to the independent set without destroying its independence. But then the independent set could not have been maximal.
- An independent set has the dominance property only if it is a maximal independent set. Thus an ***independent dominating set*** **is** the **same** as a **maximal independent set**.
- In any graph *G*, ***domination number* α(*G*) ≤ independence number *ß*(*G*)**.



Graph G

Complete graph with 5 vertices

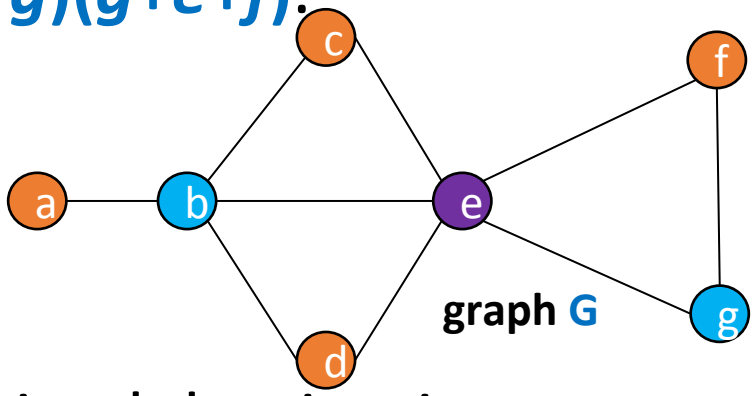


Graph H

Finding Minimal Dominating Set

- To dominate a vertex v_i we must either include v_i **or any of the vertices adjacent to** v_i. A **minimum set satisfying this condition for every vertex** v_i is a **desired set**. Therefore, **for every vertex** v_i in *G* let us form a **Boolean product of sums** $(v_i + v_{i1} + v_{i2} + \ldots + v_{id})$, where $v_{i1}, v_{i2}, \ldots, v_{id}$ are the **vertices adjacent to** v_i, and d is the **degree** of v_i.
- *For* ***graph*** G, $\theta = \Pi\,(v_i + v_{i1} + v_{i2} + \ldots v_{id})$ for all v_i **in** G. When θ is expressed as a sum of products, each term in it will represent a minimal dominating set.
- $\theta=(a+b)(b+c+d+e+a)(c+b+e)(d+b+e)(e+b+c+d+f+g)(f+e+g)(g+e+f)$.
- Since in Boolean arithmetic $(x + y)x = x$,
- $\theta = (a + b)(b + c + e)(b + d + e)(e + f + g)$

  $= ae + be + bf + bg + acdf + acdg$.

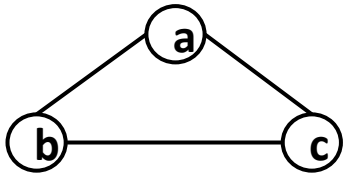


graph G

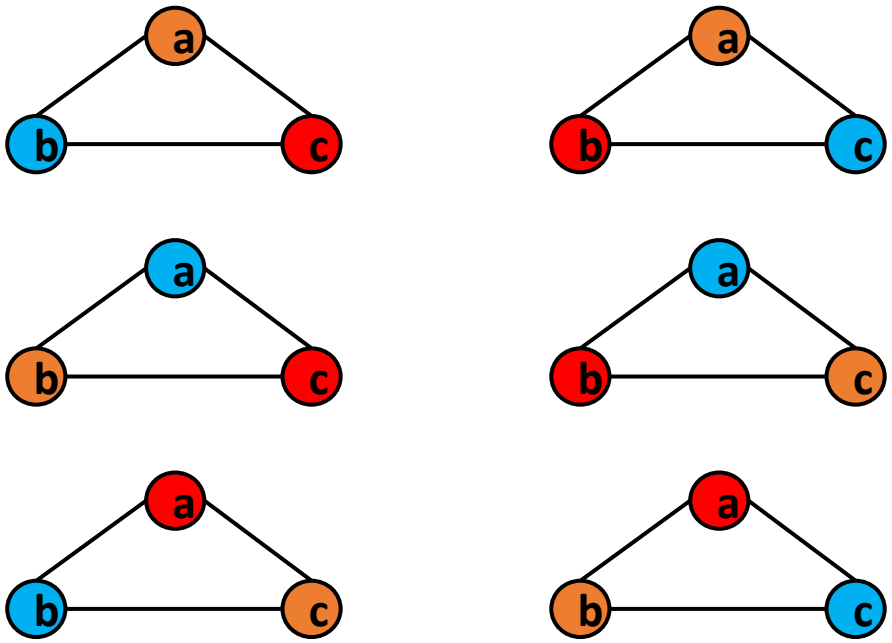
- Each of the six terms in the expression represents a minimal dominating set.

Chromatic Polynomial

- Given a **graph** with ***n*** **vertices** how many number of ways vertices of the graph can be **properly colored** (**no adjacent vertices will have same colors**) **using** λ **colors**?
  - For example, how **many different ways Graph G** with **3 vertices** can be colored using **3 colors** (🟠 🔵 🔴)?
  - 3!



graph G

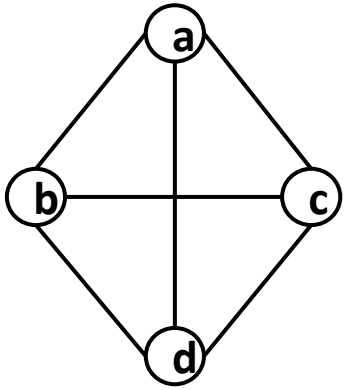


Chromatic Polynomial

- Given a **graph** with ***n*** **vertices** how many number of ways vertices of the graph can be **properly colored** (**no adjacent vertices will have same colors**) **using** ***λ*** **colors**?
- For example, the number of ways vertices of the **complete graph G** with **n**(=**4**) vertices ***G*** using **exactly** ***λ*** different colors can be found as follows:

$$=\lambda(\lambda - 1)(\lambda - 2)(\lambda - 3)$$

Vertex	No. of ways	Remark
a	λ	Out of λ colours, aanyone color can be assigned to node a
b	λ-1	Once a color is assigned to vertex a, anyone color out of λ-1 colors can be assigned to node b which is adjacent to node a.
c	λ-2	Once a color is assigned to vertices a and b, anyone color out of λ-2 colors can be assigned to node c which is adjacent to node a and b.
d	λ-3	Once a color is assigned to vertices a, b and c, anyone color out of λ-3 colors can be assigned to node d which is adjacent to node a, b and c.



graph G

Chromatic Polynomial

- The **chromatic polynomial** $P_n(\lambda)$ is defined as the number of ways of properly coloring vertices of the graph with n **vertices**, using λ **or** **fewer** **colors**. $P_n(\lambda)$ can be found as follows:
- For i = **1** to n
  - the different ways of properly coloring G using **exactly** i different colors(c_i) **x** no. of ways of choosing i colors out of λ colors ($\binom{\lambda}{i}$ or $^{\lambda}C_i$), i.e. $c_i * \binom{\lambda}{i}$.
- Since i can be any positive integer from **1** to **n** (as it is not possible to use more than **n** colors on **n** vertices), the chromatic polynomial is a sum of these terms. Therefore
- $P_n(\lambda) = \sum_{i=1}^{n} c_i * \binom{\lambda}{i}$

$$= \sum_{i=1}^{n} c_i * \frac{\lambda!}{i!(\lambda-i)!} = c_1 * \frac{\lambda}{1!} + c_2 * \frac{\lambda(\lambda-1)}{2!} + c_3 * \frac{\lambda(\lambda-1)(\lambda-2)}{3!} + \ldots + c_n * \frac{\lambda(\lambda-1)(\lambda-2)\ldots(\lambda-n+1)}{n!}$$

Chromatic Polynomial Example-1

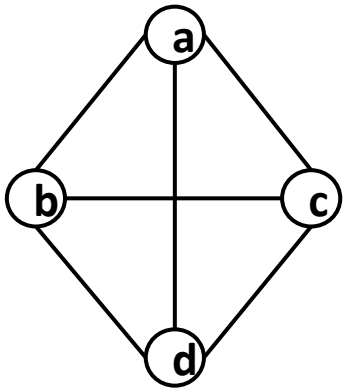
- For n=4, $P_n(\lambda)$ can be **calculated** as
  - no. of ways of coloring using exactly **1** color * ${}^{\lambda}C_1$ **+** no. of ways of coloring using exactly **2** color * ${}^{\lambda}C_2$ **+** no. of ways of coloring using exactly **3** color * ${}^{\lambda}C_3$ **+** no. of ways of coloring using exactly **4** color * ${}^{\lambda}C_4$

    $= 0 * {}^{\lambda}C_1 + 0 * {}^{\lambda}C_2 + 0 * {}^{\lambda}C_3 + 4! * {}^{\lambda}C_4$

    $= 4! \text{ x } {}^{\lambda}C_4$

    $= 4! \text{ x } \frac{\lambda!}{4!\text{x}(\lambda-4)!}$

    $= \lambda(\lambda - 1)(\lambda - 2)(\lambda - 3)$



graph G

- Therefore, for graph **G**, $P_4(\lambda) = \lambda(\lambda - 1)(\lambda - 2)(\lambda - 3)$

Chromatic Polynomial Example-2

- For this graph with n=3, $P_3(\lambda)$ can be **calculated** as
  - no. of ways of coloring using exactly **1** color * ${}^{\lambda}C_1$ **+** no. of ways of coloring using exactly **2** color * ${}^{\lambda}C_2$ **+** no. of ways of coloring using exactly **3** color * ${}^{\lambda}C_3$

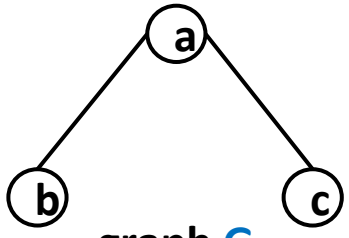
    $= 0 * {}^{\lambda}C_1 + 2 * {}^{\lambda}C_2 + 3! * {}^{\lambda}C_3$

    $= 2 * \frac{\lambda!}{2!x(\lambda-2)!} + 3! * \frac{\lambda!}{3!x(\lambda-3)!}$

    $= \lambda(\lambda - 1) + \lambda(\lambda - 1)(\lambda - 2)$

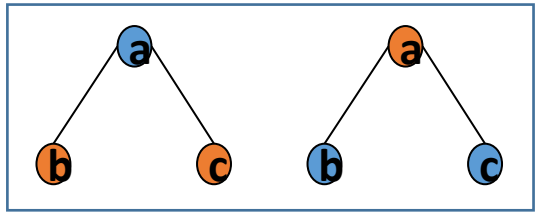
    $= \lambda(\lambda - 1)(1 + \lambda - 2)$

    $= \lambda(\lambda - 1)(\lambda - 1) = \lambda(\lambda - 1)^2$



graph G

- Therefore, for graph **G**, $\mathbf{P_3(\lambda) = \lambda(\lambda - 1)^2}$
- Note: no. of ways of coloring using exactly 1 color = 0, coloring not possible
- Note: no. of ways of coloring using exactly 2 color = 2
- Note: no. of ways of coloring using exactly 3 color = 3!



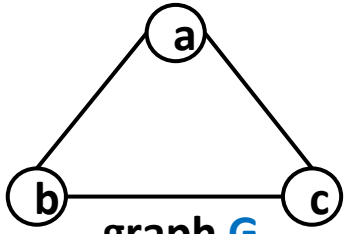
Chromatic Polynomial Example-3

- For this graph with n=3, $P_3(\lambda)$ can be **calculated** as
  - no. of ways of coloring using exactly **1** color * ${}^{\lambda}C_1$ **+** no. of ways of coloring using exactly **2** color * ${}^{\lambda}C_2$ **+** no. of ways of coloring using exactly **3** color * ${}^{\lambda}C_3$

    $= 0 * {}^{\lambda}C_1 + 0 * {}^{\lambda}C_2 + 3! * {}^{\lambda}C_3$

    $= 3! * \frac{\lambda!}{3!x(\lambda-3)!}$

    $= \lambda(\lambda - 1)(\lambda - 2)$



graph G

- Therefore, for graph **G**, $P_3(\lambda) = \lambda(\lambda - 1)(\lambda - 2)$
- Note: no. of ways of coloring using exactly 1 color = 0, coloring not possible
- Note: no. of ways of coloring using exactly 2 color = 0, coloring not possible
- Note: no. of ways of coloring using exactly 3 color = 3!

Chromatic Polynomial Example-4

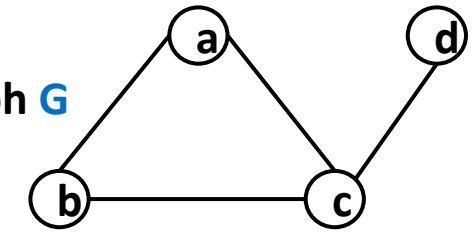
- For this graph with n=4, $P_4(\lambda)$ can be **calculated** as
  - no. of ways of coloring using exactly **1** color * $^{\lambda}C_1$ **+** no. of ways of coloring using exactly **2** color * $^{\lambda}C_2$ **+** no. of ways of coloring using exactly **3** color * $^{\lambda}C_3$ **+** no. of ways of coloring using exactly **4** color * $^{\lambda}C_4$

    $= 0 * {}^{\lambda}C_1 + 0 * {}^{\lambda}C_2 + 3!*2 * {}^{\lambda}C_3 + 4! * {}^{\lambda}C_4$

    $= 3! * 2 * \frac{\lambda!}{3!x(\lambda-3)!} + 4! * \frac{\lambda!}{4!x(\lambda-4)!}$

    $= 2*\lambda(\lambda - 1)(\lambda - 2)+ \lambda(\lambda - 1)(\lambda - 2)(\lambda - 3)= \lambda(\lambda - 1)(\lambda - 2)(2+ \lambda - 3) = \lambda(\lambda - 1)^2(\lambda - 2)$

graph G



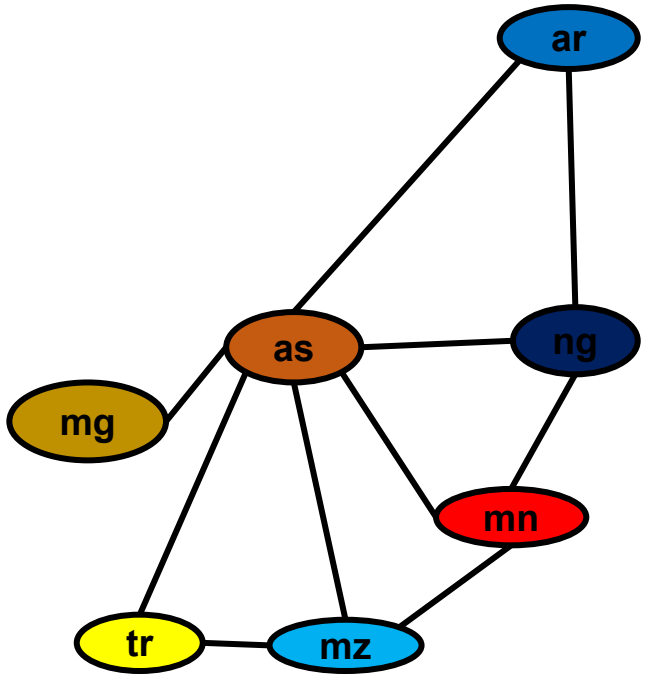
- Therefore, for graph **G**, $P_4(\lambda) = \lambda(\lambda - 1)^2(\lambda - 2)$
- Note:
  - Not possible for coloring using exactly 1 and 2 colors, because of a subgraph of 3 nodes.
  - Using 3 colors, no. of possible ways of coloring can be calculated as follows:
    - Nodes a, b, and c can be colored in **3!** Ways. Once a, b, and c are colored, d can be assigned a color which is not assigned to node c in **2** different ways. Therefore total possible ways is **3! * 2**
  - no. of ways of coloring using exactly 4 color = 4!

Theorems on Chromatic Polynomial

- For a complete graph of n vertices the chromatic polynomial is
  - $P_n(\lambda) = \lambda(\lambda - 1)(\lambda - 2) \ldots (\lambda - n + 1)$
- For a tree of n vertices the chromatic polynomial is
  - $P_n(\lambda) = \lambda(\lambda - 1)^{n - 1}$

Coloring regions of a Graph

- **What** is the **minimum number of colors** necessary to color regions of a map so that no two neighboring regions have same color?



Can we color the graph with <7 colors such as 6/5/4?

6-Color Theorem

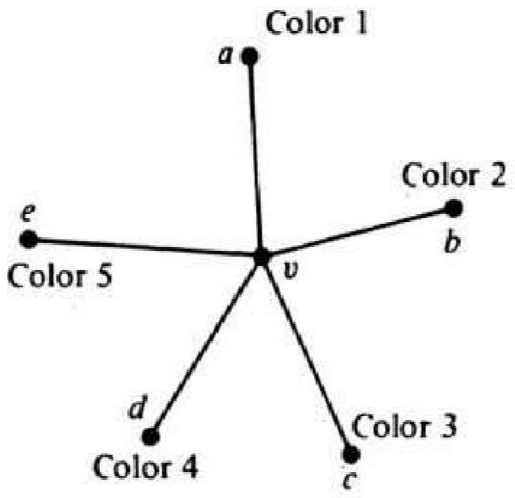
- **Lemma**: For any **simple planar graph G**, the **average degree** (**d**) of **G** is **strictly less than 6**.
- **Proof**:
  - In a graph, $d = 2e/n$.
  - $d \leq 2(3n - 6)/n$ [in a planar graph, we have the inequality $e \leq 3n - 6$ (for $n \geq 3$)
  - $d \leq 6 - 12/n$
  - $d < 6$ [for $n \geq 3$. For $n < 3$ we can check directly.
- **Theorem**: Any **simple planar graph** can be **properly colored with six colors**.
- **Proof** (by induction on the number of vertices):
  - (**Base case**) Suppose **we have a graph** such that **$n \leq 6$**. For $n \leq 6$, we **can give each vertex a different color** and **use ≤ 6 colors**.
  - (**Induction hypothesis**) Now **assume** that **any simple planar graph** with **n** vertices **can be properly colored** with **six colors**.
  - Let **G** be any **simple planar graph** with **n+1 vertices**. From lemma above, we know that **G must have some vertex w** of **degree ≤ 5**. Remove **w** from **G to form G′**. **G′** has **n vertices** and we may **apply our induction hypothesis** to know it **can be properly colored** in **6 colors**. Therefore, **G′ can be colored with ≤ 6 colors**. Now, we can think of **this as coloring all of G except w**. But, **since w has degree at most 5, one of the 6 colors will not be used for any of the neighbors of w** and we can **finish coloring G**.

5-Color Theorem

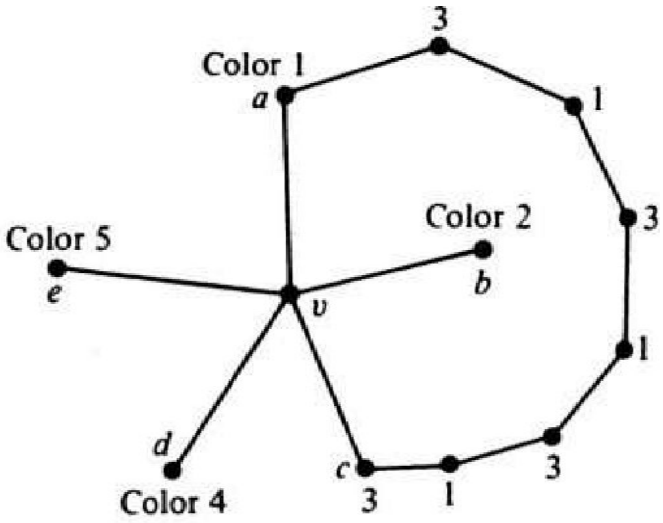
- **Theorem**: The vertices of every planar graph can be properly colored with five colors.
- **Proof** (by induction):
  - Since the vertices of **all graphs** (self-loop-free) **with 1,2,3,4, or 5 vertices** can be **properly colored with five colors**, let us **assume that vertices of every planar graph** with **n – 1 vertices** can be **properly colored** with **five colors**. Then, **if** we **prove** that **any planar graph G** with **n vertices** will **require no more than five colors**, we can **prove the theorem**.
  - Consider the **planar graph G** with **n vertices**. Since **G** is planar, it must have at least one vertex with degree five or less (**Proof in next slide**). Let **this vertex** be **v**.
  - Let G′ be a graph (of n – 1 vertices) obtained from G by deleting **vertex v** (i.e., **v and all edges incident on v**). Graph G′ requires no more than five colors, according to the induction hypothesis. Suppose that the vertices in G′ have been properly colored, and now we add to it v and all edges incident on v. If the degree of v is 1, 2, 3, or 4, we have no difficulty in assigning a proper color to v.

5-Color Theorem contd…

- **Theorem**: The vertices of every planar graph can be properly colored with five colors.
- **Proof** (by induction) contd…:
  - This leaves only the case in which the degree of v is five, and all the five colors have been used in coloring the vertices adjacent to v, as shown in Fig 1.
  - Suppose that there is a path in G′ between vertices a and c colored alternately with colors 1 and 3, as shown in Fig. 2. Then a similar path between b and d, colored alternately with colors 2 and 4, cannot exist; otherwise, these two paths will intersect and cause G to be nonplanar.
  - If there is no path between b and d colored alternately with colors 2 and 4, starting from vertex b we can interchange colors 2 and 4 of all vertices connected to b through vertices of alternating colors 2 and 4. This interchange will paint vertex b with color 4 and yet keep G′ properly colored. Since vertex d is still with color 4, we have color 2 left over with which to paint vertex v.
  - Had we assumed that there was no path between a and c of vertices painted alternately with colors 1 and 3, we would have released color 1 or 3 instead of color 2. And thus the theorem.



(a)



(b)

4-Color problem

- The most famous conjecture in graph theory is that every map (i.e., a planar graph) can be properly colored with four colors. The four-color conjecture, has been worked on by many famous mathematicians for the past 100 years. No one has yet been able to either prove the theorem or come up with a map (in a plane) that requires more than four colors.

Number of Regions in a Planar Graph

- **Theorem**: A **connected planar graph** with **n vertices** and **e edges** has **e − n + 2 regions** (**Euler's Formula**)
- Proof:
  - any planar graph can be drawn as a polygon net such that each region is a polygon with straight line edges(Theorem 1).
  - Let the polygonal net consist of ***f* regions** or faces, and let k_p be the number of **p-sided regions**. Since each edge is on the boundary of exactly two regions, we have
    - $3.k_3+4.k_4+5.k_5+\ldots+r.k_r=2.e$, where k_r is the number of polygons, with maximum edges. **Eqn. - 1**
    - Also $k_3+k_4+k_5+\ldots+k_r= f$ **Eqn. - 2**
    - The **sum of all angles** subtended **at each vertex** in the polygonal net is $2\pi n$ **Eqn. – 3**
    - The sum of all interior angles of a p-sided polygon is π(p − 2), and the sum of the exterior angles is π(p + 2), let us compute the expression in (**Eqn. - 1**) as the grand sum of all interior angles of f − 1 finite regions plus the sum of the exterior angles of the polygon defining the infinite region. This sum is
      - $\pi(3-2).k_3+\pi(4-2).k_4+\pi(5-2).k_5+\ldots+\pi(r-2).k_r+4\pi = \pi(2e-2f)+4\pi$ **Eqn. – 4**
    - **From eqn 3 and 4 we have $2\pi(e - f) + 4\pi = 2\pi n$, or $e - f + 2 = n$.**
    - Therefore, the number of regions is f = e − n + 2.
- **Note**:
  - It is sufficient to prove the theorem for a simple graph, because adding a self-loop or a parallel edge simply adds one region to the graph and simultaneously increases the value of e by one. We can also remove all edges that do not form boundaries of any region. Addition (or removal) of any such edge increases (or decreases) e by one and increases (or decreases) n by one, keeping the quantity e − n unaltered.